

1.0 Introduction

Network design and implementation is considered an important aspect in every business that tend to connect its stakeholders, business, products etc. for better leads in the market demand position and horizon since it enables and define the communication of every networking devices in that organization, and what resources is communicated to what location and to which audiences. Therefore, when designing and implementing, it should be properly executed, managed and maintained considering all the best design practices. For example, as according to Azzedin & Maheswaran (2012), the network should be of high performance and secured adhering to information security principles of CIA (Confidentiality, Integrity and Availability) in ensuring that to the network is secured, accountable and reliable. The purpose of this study is to analyse the DFS enterprise network requirements, design and implement the desired network while giving a detailed documentation on the design and implementation strategies.

2.0 Network Design Principles

According to Hamid (2015), there is a minimal chance of a network being deemed perfect if it is subjected to expert's evaluation and penetration capabilities of attackers. Therefore, when design a network, engineers should not be too confident and reluctant on their job but to sort out for other means of making their network adhere to information security principles of CIA (Confidentiality, Integrity and Availability), these includes to measures like disaster recovery, updating to new technologies, making regular backups among other network design principles or preventive measures. As stated by Vestin et. al (2015), network design principles define the network design goals, choices, approaches, process, capacity planning, desired requirements among others.

Therefore, the network design principles that should be included in the best network design practices includes;

2.1. Hierarchical Modularity and Performance

Building a network using a hierarchical model enables separate networking segments to be networked together and it gives three conceptual layers of the network for improved performance, security, and easy maintenance (Danilo, 2018). Network hierarchical model together with mesh topology includes all nodes cooperating to distribute data amongst each other. These conceptual layers, including core, distribution, and access, provide different functionalities networked to achieve one specific network goal. Sinket (2019) explains that the hierarchical network model

provides a framework that network designers can use to help ensure that the network is flexible and easy to implement and troubleshoot.

Considering the OSI model, the hierarchical model is seen at all the OSI layers starting from the physical, data-link, and network among other layers. This explicitly explains the network performance at every OSI layer

Therefore, this is a very important aspect of design consideration that should be indispensable at every network design and implementation, as explained above, the hierarchical model achieves improved performance, improved security, high network reliability, and easy management.

2.2. Redundancy and Scalability

To provide availability and reliability of the network, the network design should be designed to prevent a single point of failure. Mas-Machuca (2016), explains redundancy as a networking strategy of having multiple traffic paths such that whenever a link gets down, the communication will still carry on with other alternative links since there are redundant links. This report further suggests that there should be redundancy in the network, so that a single link or hardware failure does not isolate any portion of the network resulting in those users losing access to network resources. The amount of redundancy required varies from network to network. Some networks might require a backup link between two sites, and some networks might require redundant links, routers, and switches.

Redundancy breed scalability which is the ability to design a flexible network that can accommodate future expansion or growth, a network that that additional nodes can be added without affecting the network performance and operation ability.

In the OSI model, redundancy starts from the physical layer of having multiple links that traffic can flow through such that when one link fails, the traffic will be directed through the alternative paths. At the data link layer, redundancy is implemented through protocols such as Spanning Tree Protocol (STP) and EtherChannel, STP will prevent any possible loops in the network while providing redundancy when one link fails. Similarly, EtherChannel will bundle together multiple links hence enhancing bandwidth and redundancy in case of a single point of failure in the channel. At the network layer of the OSI model, several protocols should be implemented to enhance redundancy, Hot Stand-by Routing Protocols and Virtual Router Redundancy Protocols are the protocols that are responsible for providing redundancy at layer 3 in case one node fails.

Therefore, redundancy is a very important network design principle that offers to provide a backup path as an alternative to the primary path so that if the primary path fails, traffic will automatically run across the backup path. This is an expert evaluation is a broader foundation for network

scalability (Sinket, 2019). To expand the network, we would need a scalable network that will allow us to add more nodes without affecting the network performance and operation ability, therefore redundancy goes hand in hand with scalability.

2.3. Security

Security is one of the most important factors in every network design and implementation strategy. A good network should be able to implement the security policy based on the principle of CIA (Confidentiality, Integrity, and Availability) to ensure the network is secured, accountable and reliable. Protecting data at rest and in use is a very vital aspect of security that the enterprise network should look into to ensure that the data is restricted by user role and limiting system access to those who the data is intended using AAA technique. The network should be protected against malware attacks such as ransomware attacks, viruses, worms and among others, and finally, there should be continuous network management to ensure better performance.

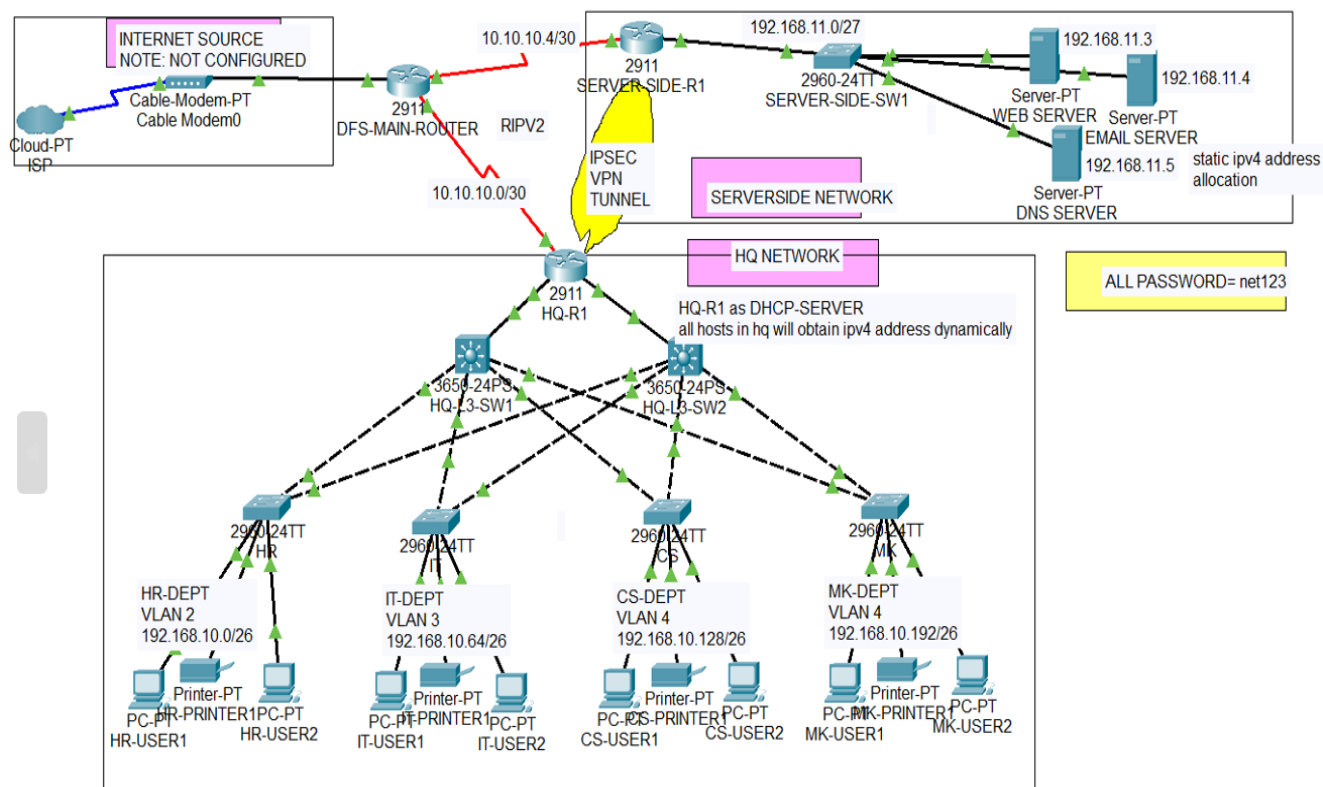
At the physical layer of the OSI model, network security starts from the physical security of how the devices are stored and managed, how they are protected from human intrusion or natural disasters. The physical security ensures that the networking devices are safe and free from any inconveniences or disasters. At the data-link layer, there are some of the protocols that can be implemented to enhance security at the network. Virtual Local Area Network (VLAN) and Port-security are some of the important protocols that ensure security at the data-link layer. VLANs will separate the broadcast domain and make each department operate independently hence no user from different VLAN can infiltrate to other departments. Port-security will ensure that a port only accepts the assigned host device and all intruders cannot access the network. At the network layer of OSI model, firewall works to provide security to the network. The firewall will filter all the traffic coming and leaving the network such that all the unwanted one are blocked and the desired one are allowed in the network based on the firewall rules.

3.0 Design

3.1. Hierarchical Network Design Model (LAN)

As the best design strategy, hierarchical network model was implemented on DFS network having the core layer, distribution layer and access layer as shown below, and also the topology used in the design is the mesh topology on the HQ part, which into a brief description includes all nodes cooperating to distribute data amongst each other.

The following attached is the proposed network design;



3.2. Subnetting (Design IP Addressing and Allocation)

During IP address allocation, we evaluated the number of subnets involved in the HQ network and server-side site. Also, we allocated IP address to be used between the three routers. After deciding which IP to use we carried out subnetting to enable devices in the network to access an IP address in their respect subnet or network.

The following is the addressing table we used.

Between the Routers

No.	Routers	Network & Subnet Mask	Valid Host Addresses	Broadcast Address
1	HQ-R1 and DFS-Main-Router	10.10.10.0/30	10.10.10.1 and 10.10.10.2	10.10.10.3
2	DFS-Main-Router and Server-side-R1	10.10.10.4/30	10.10.10.5 and 10.10.10.6	10.10.10.7

HQ Departments

No.	Department	Network & Subnet Mask	Valid Host Addresses	Default Gateway	Broadcast Address
1	HR	192.168.10.0/26	192.168.10.1 to 192.168.10.62	192.168.10.1	192.168.10.63
2	IT	192.168.10.64/26	192.168.10.65 to 192.168.10.126	192.168.10.65	192.168.10.127
3	CS	192.168.1.128/26	192.168.10.129 to 192.168.10.190	192.168.10.129	192.168.10.191
4	MK	192.168.1.192/26	192.168.10.193 to 192.168.10.254	192.168.10.193	192.168.10.255

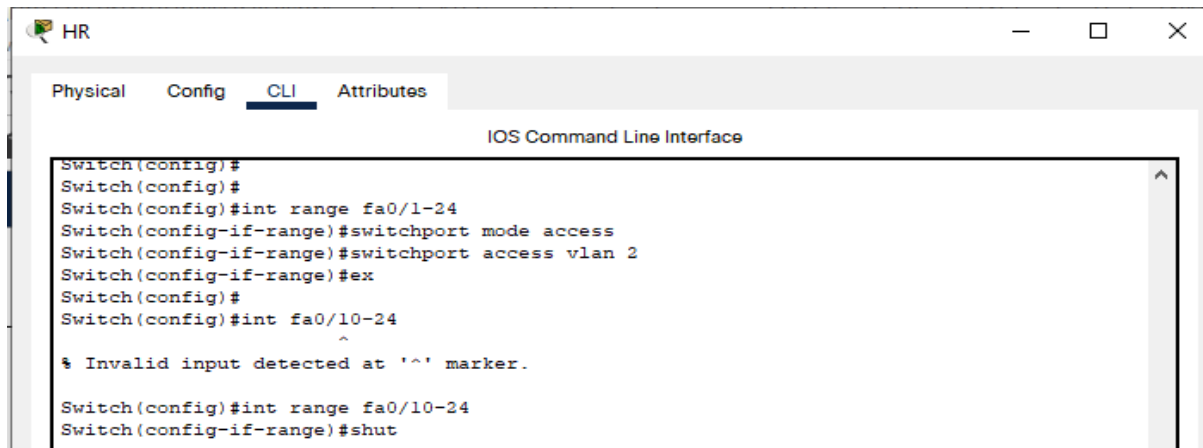
Server-Side Site

No.	Subnet Used	Network & IP Address	Valid Host Addresses	Default Gateway	Broadcast Address
1	Subnet 1	192.168.11.0/27	192.168.11.1 to 192.168.11.30	192.168.11.1	192.168.11.31

3.3. VLAN Configuration, VLAN Trunking 802.1Q, Inter-VLAN Routing

a. VLANs

For improved security and easy maintenance, each department in the HQ network is in a different VLAN and assigned to different subnet, by default the devices in different VLANs will not communicate unless inter-VLAN is implemented for communication. The following are sample of VLAN configuration commands and results in the switches.



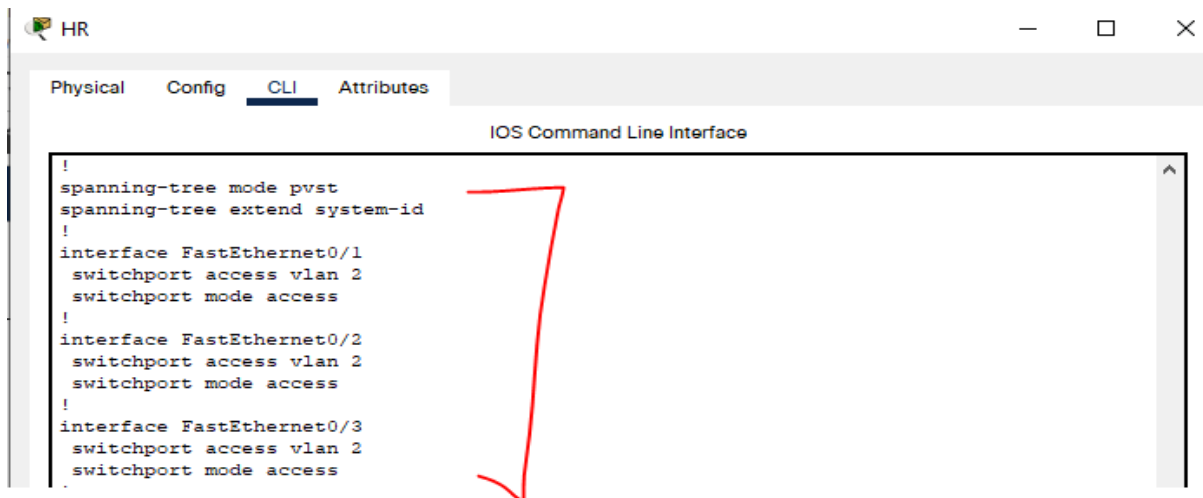
HR

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch(config)#
Switch(config)#
Switch(config)#int range fa0/1-24
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#ex
Switch(config)#
Switch(config)#int fa0/10-24
Switch(config-if-range)#shut
```

% Invalid input detected at '^' marker.



HR

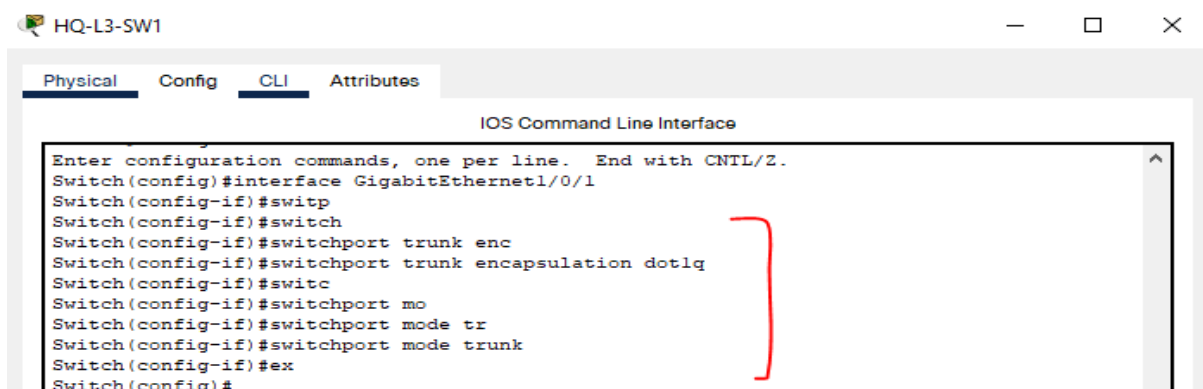
Physical Config CLI Attributes

IOS Command Line Interface

```
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 2
switchport mode access
!
interface FastEthernet0/2
switchport access vlan 2
switchport mode access
!
interface FastEthernet0/3
switchport access vlan 2
switchport mode access
.
```

b. VLAN Trunking (802.1Q)

To have VLAN traffic between the switches in the network, we used trunk links to facilitate this. A trunk link is a normal link that is able to pass traffic from different VLANs and has a method to separate traffic between VLANs. The following are sample of VLAN Trunking configuration commands and results in the switches.



HQ-L3-SW1

Physical Config CLI Attributes

IOS Command Line Interface

```
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface GigabitEthernet1/0/1
Switch(config-if)#switp
Switch(config-if)#switch
Switch(config-if)#switchport trunk enc
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switc
Switch(config-if)#switchport mo
Switch(config-if)#switchport mode tr
Switch(config-if)#switchport mode trunk
Switch(config-if)#ex
Switch(config)#
```

The screenshot shows the CLI of a switch named HQ-L3-SW1. The 'CLI' tab is selected. The user has entered the command 'do sh interface trunk' to display the configuration for all trunk ports. The output shows that Gig1/0/1 is configured as a trunk port with 802.1Q encapsulation, status 'trunking', and native VLAN 1. A red bracket highlights the 'Native vlan' column.

```
Switch(config)#do sh switchport trunk
sh switchport trunk
^
% Invalid input detected at '^' marker.

Switch(config)#do sh interface trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig1/0/1   on        802.1q         trunking     1

Port      Vlans allowed on trunk
Gig1/0/1   1-1005

Port      Vlans allowed and active in management domain
Gig1/0/1   1,2,3,4,5

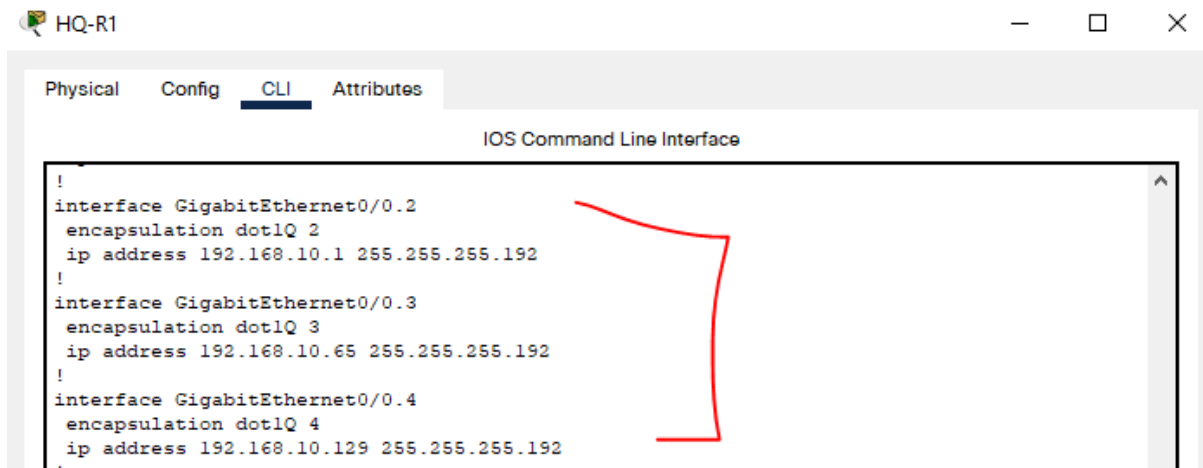
Port      Vlans in spanning tree forwarding state and not pruned
Gig1/0/1   1,2,3,4,5
```

c. Inter-VLAN Routing

By default, the devices in different VLANs will not communicate unless inter-VLAN is implemented for communication. During the configuration, we used Router-on-a-Stick as the protocol for inter-VLAN routing, in which from a single physical interface, we created several sub-interfaces and assigned required parameters such as the IEEE 802.1Q encapsulation and IP addresses. The following are sample of inter-VLAN routing configuration commands and results in the HQ-R1.

The screenshot shows the CLI of a router named HQ-R1. The 'CLI' tab is selected. The user has entered the command 'do wr' to save the configuration. The output shows the configuration for the sub-interface GigabitEthernet0/0.2, which is configured with 802.1Q encapsulation and IP address 192.168.10.1/24. A red bracket highlights the configuration lines.

```
Router(config)#
Router(config)#
Router(config)#interface GigabitEthernet0/0.2
Router(config-subif)# encapsulation dot1Q 2
Router(config-subif)# ip address 192.168.10.1 255.255.255.192
Router(config-subif)#ex
Router(config)#do wr
Building configuration...
[OK]
```



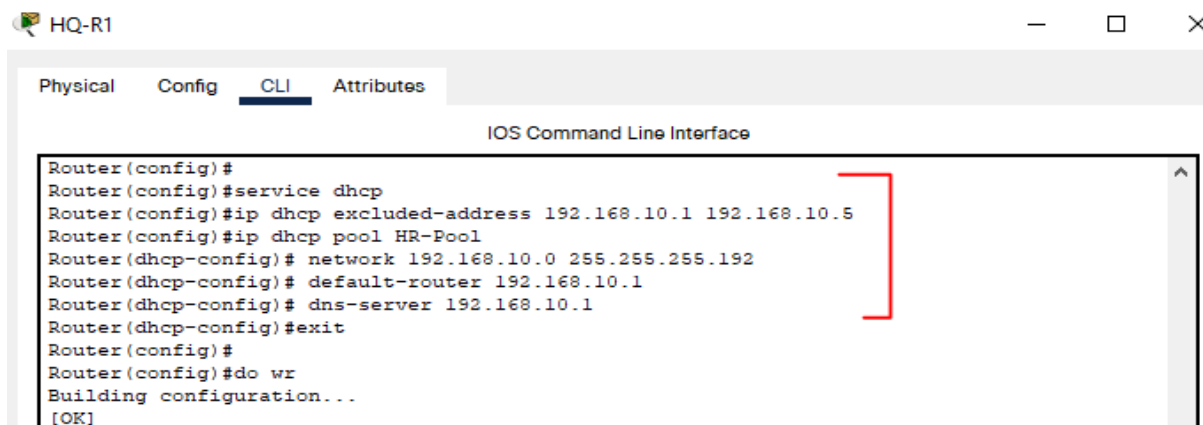
The screenshot shows the CLI of HQ-R1 with three interfaces configured. A red bracket highlights the configuration for interfaces GigabitEthernet0/0.2, 0.3, and 0.4. Each interface is configured with 'encapsulation dot1Q' and an 'ip address' from the 192.168.10.0/24 network.

```
!
interface GigabitEthernet0/0.2
 encapsulation dot1Q 2
 ip address 192.168.10.1 255.255.255.192
!
interface GigabitEthernet0/0.3
 encapsulation dot1Q 3
 ip address 192.168.10.65 255.255.255.192
!
interface GigabitEthernet0/0.4
 encapsulation dot1Q 4
 ip address 192.168.10.129 255.255.255.192
```

3.4. DHCP Server, DNS Server, Web Server, Email Server

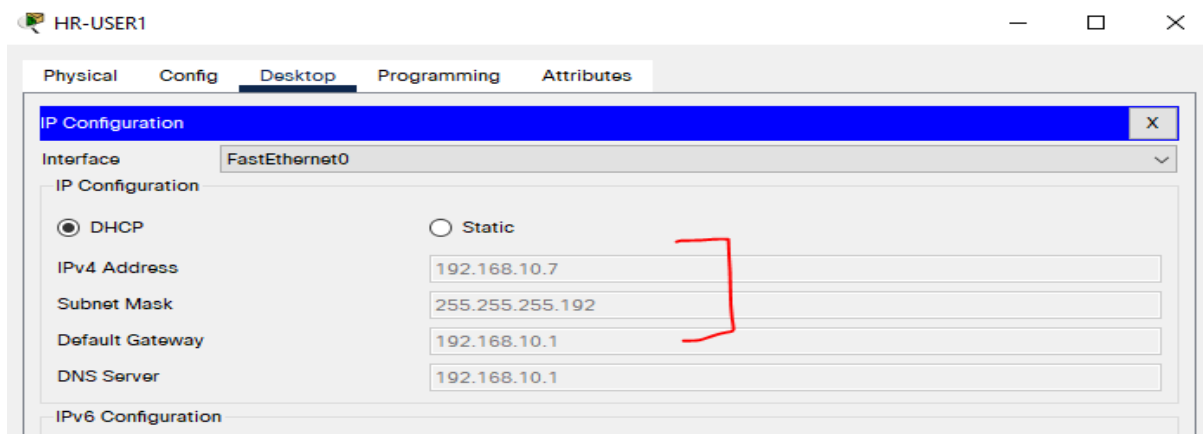
a. DHCP Server

All the host devices in the HQ-network are allocated IPv4 addresses dynamically. HQ-R1 was used as the DHCP server. The diagrams below show DHCP server configurations and an evidence of automatic IPv4 assignment.



The screenshot shows the CLI of HQ-R1 with the DHCP service configured. A red bracket highlights the configuration for the DHCP pool 'HR-Pool'. The configuration includes setting the network, excluded address, default router, and DNS server.

```
Router(config)#
Router(config)#service dhcp
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.5
Router(config)#ip dhcp pool HR-Pool
Router(dhcp-config)# network 192.168.10.0 255.255.255.192
Router(dhcp-config)# default-router 192.168.10.1
Router(dhcp-config)# dns-server 192.168.10.1
Router(dhcp-config)#exit
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
```

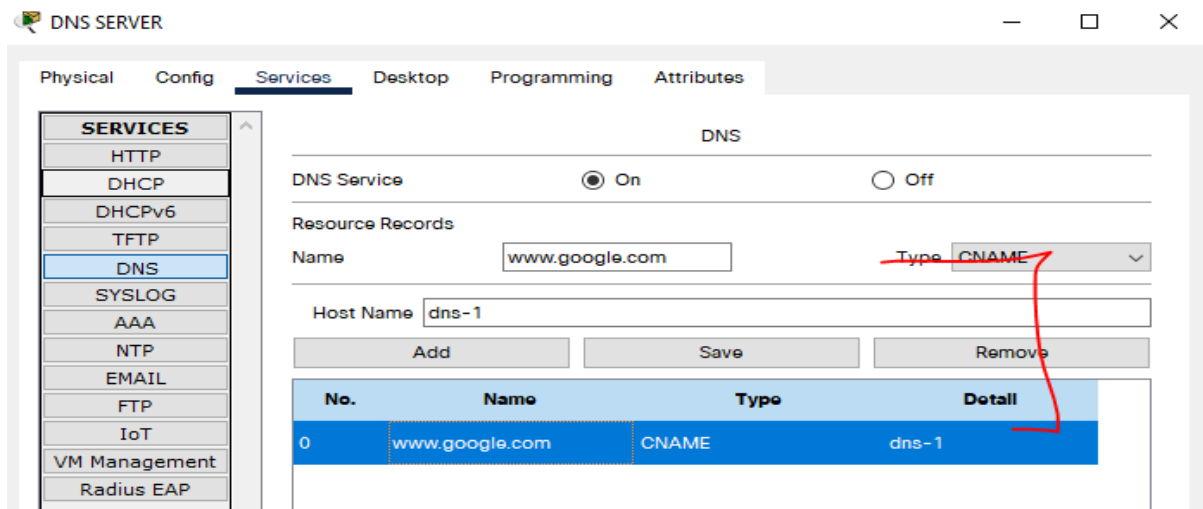


The screenshot shows the Desktop IP Configuration window for HR-USER1. The 'FastEthernet0' interface is selected, and the 'DHCP' option is chosen. A red bracket highlights the IP configuration fields: IPv4 Address (192.168.10.7), Subnet Mask (255.255.255.192), Default Gateway (192.168.10.1), and DNS Server (192.168.10.1).

Interface	FastEthernet0
IP Configuration	
☒ DHCP	☐ Static
IPv4 Address	192.168.10.7
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1
DNS Server	192.168.10.1
IPv6 Configuration	

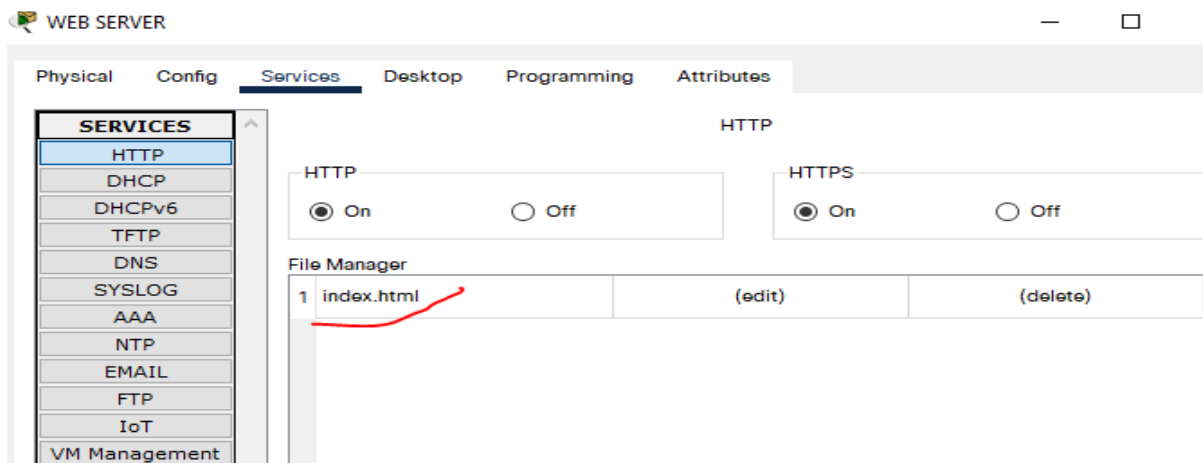
b. DNS Server

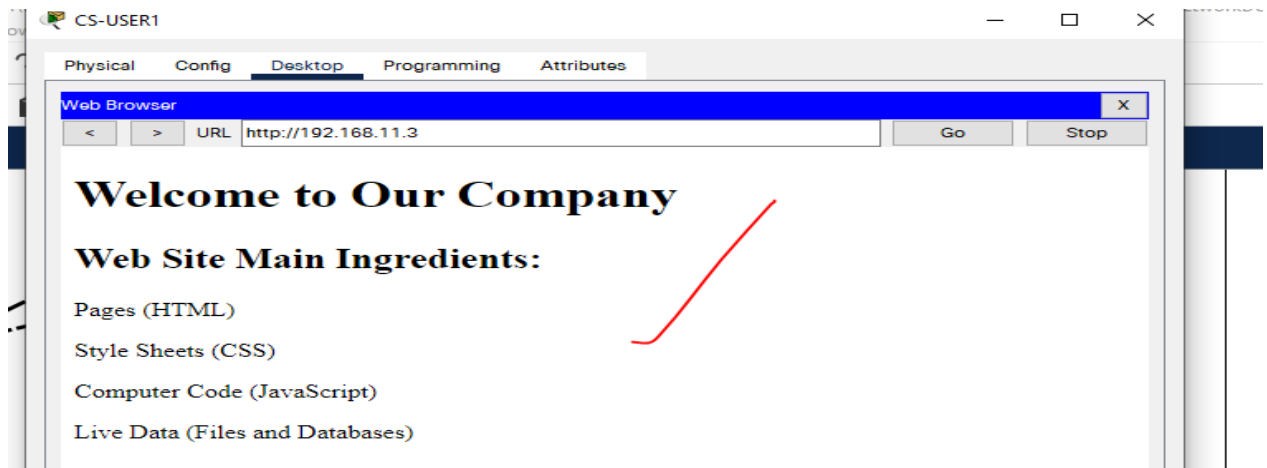
For domain name resolution, DNS server was used in the network with the below configuration.



c. Web Server

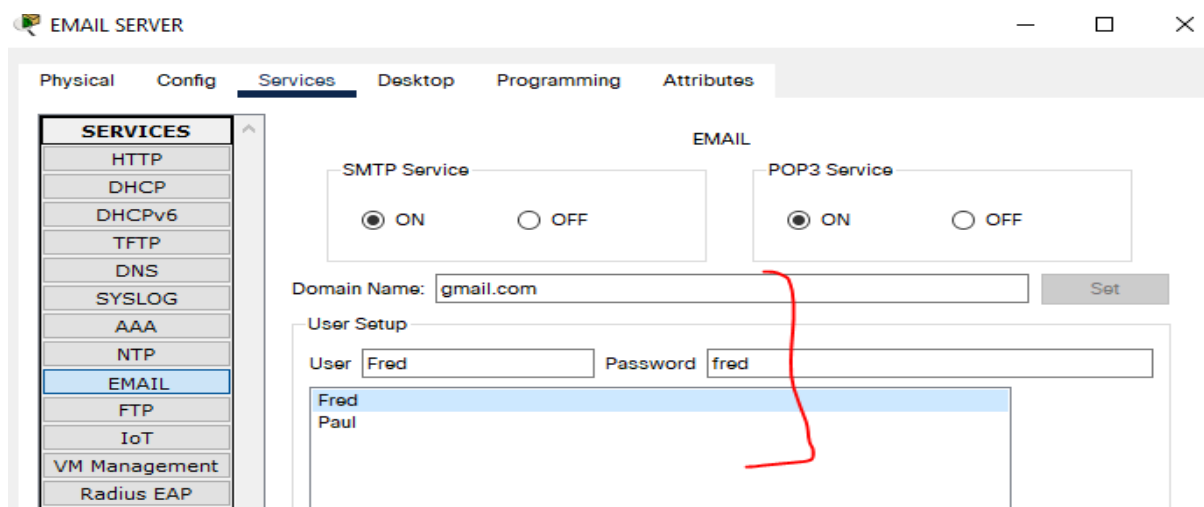
The web server in the network was configured with other html files as per our need and all the hosts can access the web pages. The following are sample of the configuration and results in the hosts devices.

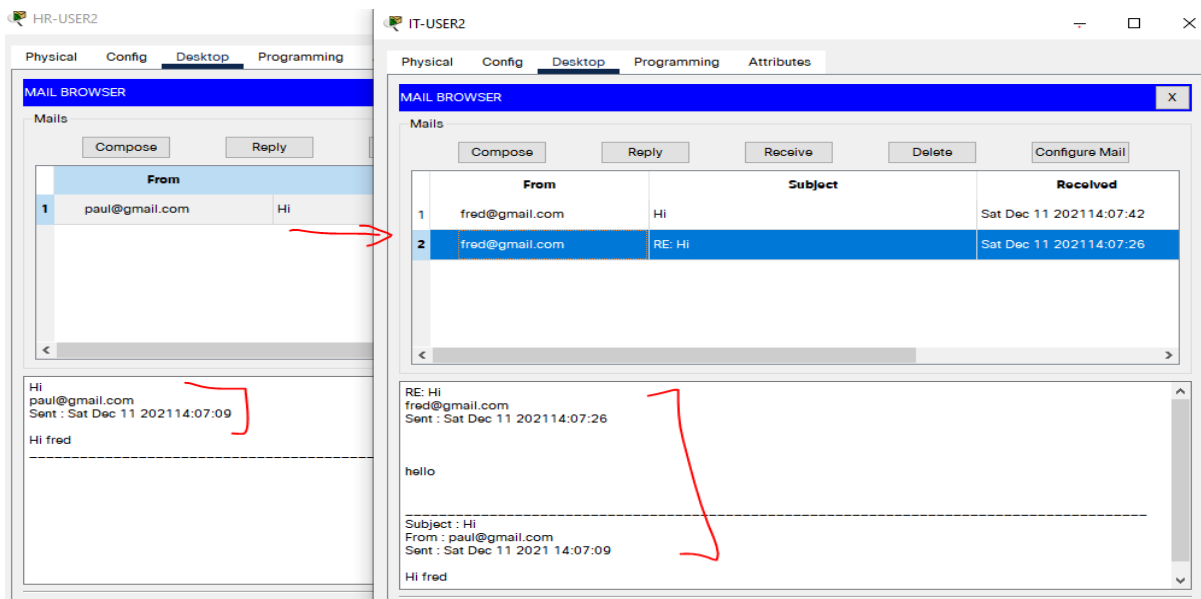




d. Email Sever

To enable sending and receiving of mails in the network using SMTP and POP3 protocols, the email server was configured to facilitate this process. The following are sample of the configuration and results in the hosts devices.





3.5. Dynamic Routing Protocol Configuration using RIPv2

RIPv2 was used as routing protocol to advertise routes in the network and enable communication in the network. The following shows RIPv2 configuration commands and output in the router.



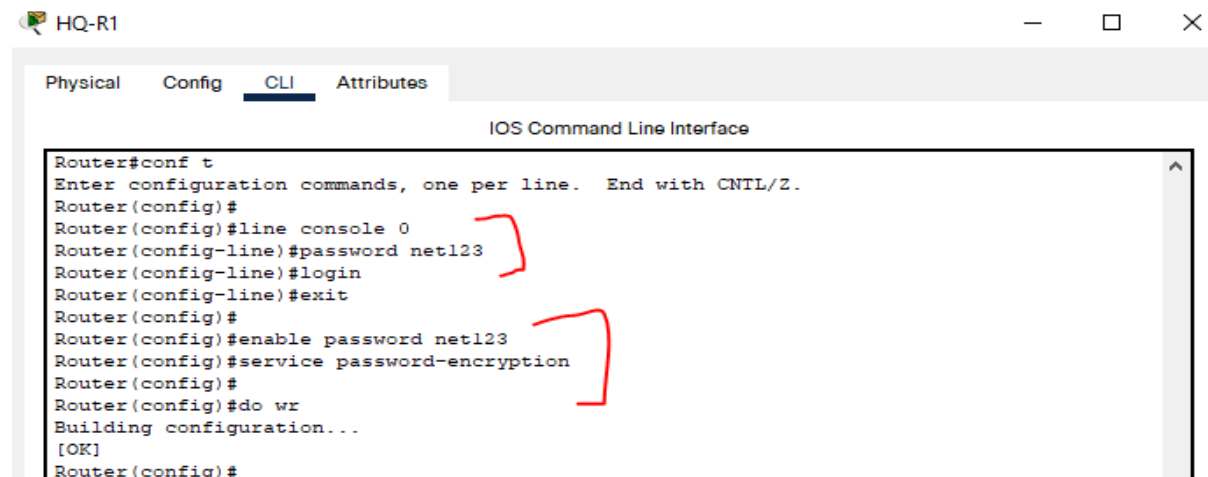
4.0 Access Control, User Privileges and IPsec VPN

a. User Privileges

Security is a very important milestone in network design and implementation. Therefore, users' privileges were implemented both in line console user and privileged exec mode. Both line console and privileged exec mode are protected using the passwords and all these passwords are encrypted. The console port is mainly used for local system access using a console terminal. By default, the console port does not require a password for console administrative access. However, you should

always configure a console port line-level password. The enable password is used for securing privilege mode. This password will be shown in clear text by the command “show running-configuration” unless the service password-encryption password is used.

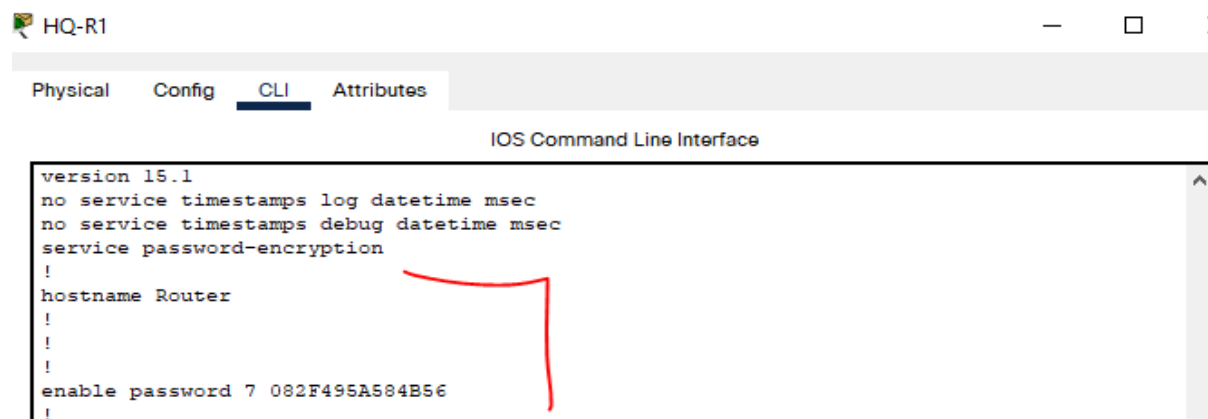
The diagrams show a sample configuration commands and results in the devices in the network.



Physical Config **CLI** Attributes

IOS Command Line Interface

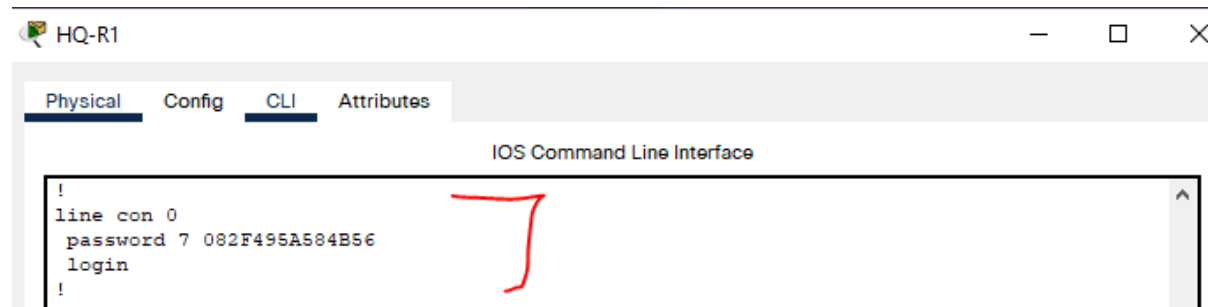
```
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#line console 0
Router(config-line)#password net123
Router(config-line)#login
Router(config-line)#exit
Router(config)#
Router(config)#enable password net123
Router(config)#service password-encryption
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```



Physical Config **CLI** Attributes

IOS Command Line Interface

```
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
service password-encryption
!
hostname Router
!
!
!
enable password 7 082F495A584B56
!
```



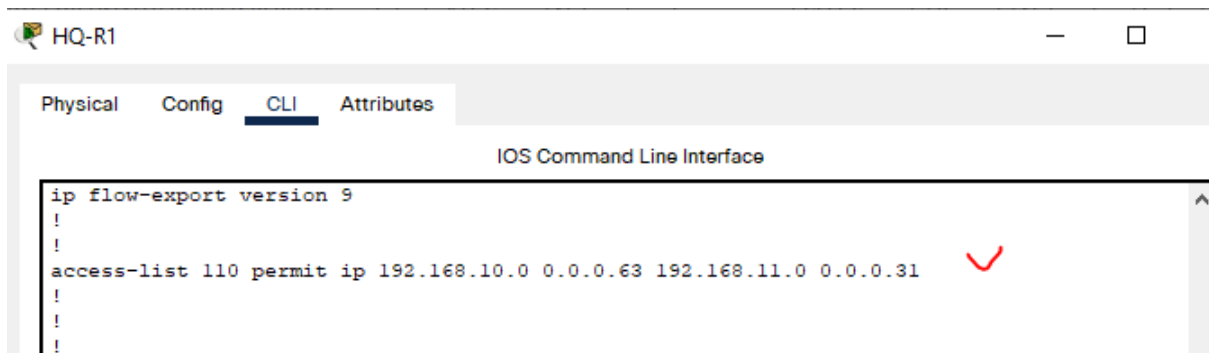
Physical Config **CLI** Attributes

IOS Command Line Interface

```
!
line con 0
password 7 082F495A584B56
login
!
```

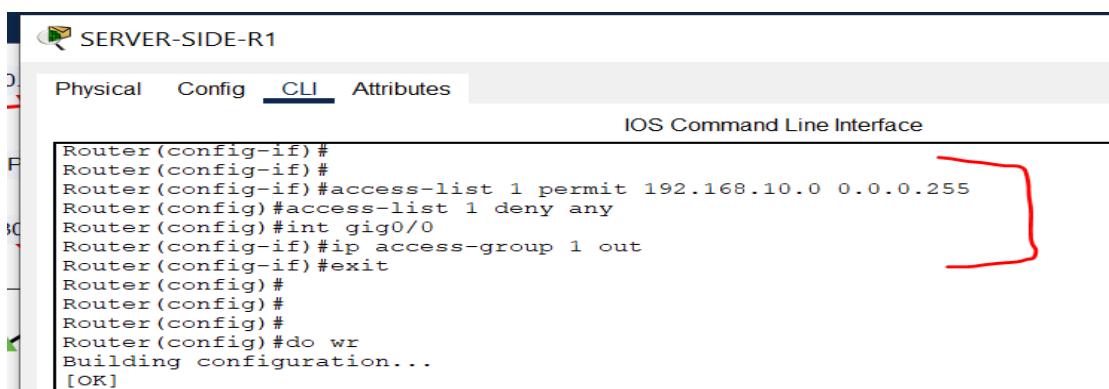
b. Access Control Lists

According to Pluralsight (2019), ACLs are a network filter utilized by routers and some switches to permit and restrict data flows into and out of network interfaces. When an ACL is configured on an interface, the network device analyzes data passing through the interface, compares it to the criteria described in the ACL, and either permits the data to flow or prohibits it. To enhance security, ACLs were used to filter traffic based on the set of rules set. To implement secure communication between HR department and the servers, the first access control list was implemented together with site-to-site VPN in HQ and Server-side routers. The second ACL was implemented in server-side R1 to allow only HQ departments to access the servers while blocking any other network from accessing the servers. The diagrams show a sample configuration commands and results in the devices in the network.



The screenshot shows the CLI interface of HQ-R1. The 'CLI' tab is selected. The command 'ip flow-export version 9' is entered. Below it, an access list is configured: 'access-list 110 permit ip 192.168.10.0 0.0.0.63 192.168.11.0 0.0.0.31'. A red checkmark is placed to the right of the command line.

```
ip flow-export version 9
!
!
access-list 110 permit ip 192.168.10.0 0.0.0.63 192.168.11.0 0.0.0.31
!
!
```



The screenshot shows the CLI interface of SERVER-SIDE-R1. The 'CLI' tab is selected. The following commands are entered: 'Router(config-if)#', 'Router(config-if)#', 'Router(config-if)#access-list 1 permit 192.168.10.0 0.0.0.255', 'Router(config)#access-list 1 deny any', 'Router(config)#int gig0/0', 'Router(config-if)#ip access-group 1 out', 'Router(config-if)#exit', 'Router(config)#', 'Router(config)#', 'Router(config)#', 'Router(config)#do wr', 'Building configuration...', and '[OK]'. A red bracket is drawn around the three lines: 'Router(config-if)#access-list 1 permit 192.168.10.0 0.0.0.255', 'Router(config)#access-list 1 deny any', and 'Router(config-if)#ip access-group 1 out'.

```
Router(config-if)#
Router(config-if)#
Router(config-if)#access-list 1 permit 192.168.10.0 0.0.0.255
Router(config)#access-list 1 deny any
Router(config)#int gig0/0
Router(config-if)#ip access-group 1 out
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#do wr
Building configuration...
[OK]
```

c. IPsec VPN

According to GeeksforGeeks (2020), Virtual Private Networks or VPNs allows network users to access a private network over the Internet securely and privately. The VPN technology will tend to create an encrypted connection called VPN tunnel, and all Internet traffic and communication is passed through this secure tunnel. Therefore, the Internet Protocol Security IPsec VPN will provide secure internet communication over the IP network. IPsec will secure the IP communication by verifying the session and encrypts all data packet during the communication.

For security purposes in our network, a site-to-site VPN was implemented between the HQ and server-side routers to enable HR department to have secure communication between HR department and the servers. This will ensure that all traffic between the HR department and the servers are encrypted and cannot be read by sniffers. The implemented ACL species a rule of allowing only HR subnet to have the secure communication with the servers.

The diagrams below show a sample of IPsec VPN configuration plus an indication of unencrypted and encrypted traffic.

```

HQ-R1
Physical Config CLI Attributes
IOS Command Line Interface
!
license udi pid CISCO2911/K9 sn FTX1524BKMH-
license boot module c2900 technology-package securityk9
!
!
!
crypto isakmp policy 10
  encr aes 256
  authentication pre-share
  group 5
!
crypto isakmp key vpnpa55 address 10.10.10.6
!
!
crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
!
crypto map VPN-MAP 10 ipsec-isakmp
  description VPN connection to SS
  set peer 10.10.10.6
  set transform-set VPN-SET
  match address 110
!

```

```

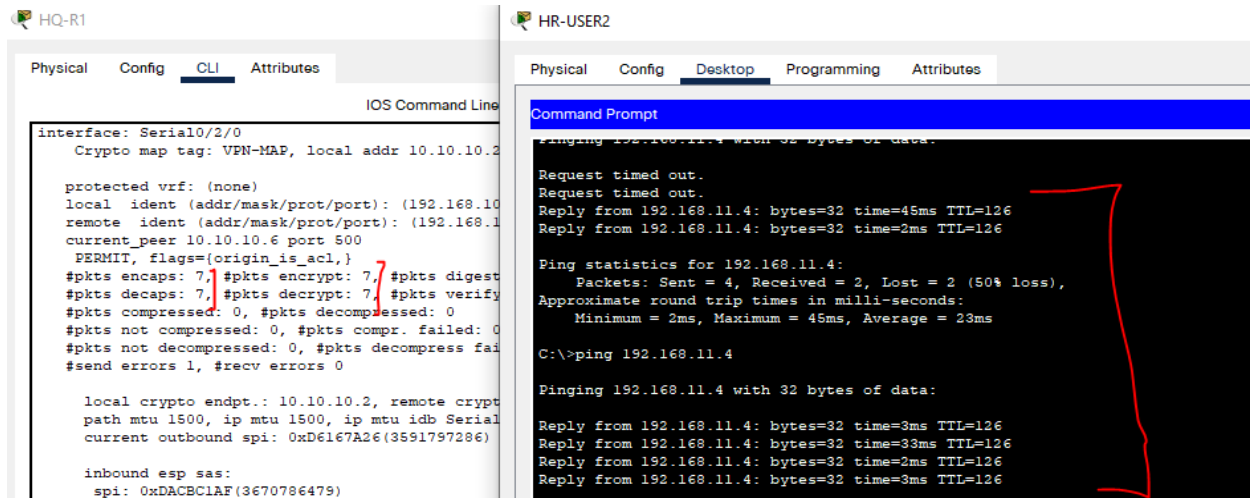
Router#sh crypto ip
Router#sh crypto ipsec s
interface: Serial0/2/0
  Crypto map tag: VPN-MAP, local addr 10.10.10.2
protected vrf: (none)
local ident (addr/mask/prot/port): (192.168.10.0/255.255.0.0/0/0)
remote ident (addr/mask/prot/port): (192.168.11.0/255.255.0.0/0/0)
current_peer 10.10.10.6 port 500
  PERMIT, flags={origin_is_acl,}
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0
local crypto endpt.: 10.10.10.2, remote crypto endpt.: 10.10.10.6
path mtu 1500, ip mtu 1500, ip mtu idb Serial0/2/0: 1500
current outbound spi: 0x0(0)
inbound esp sas:
inbound ah sas:

```

```

IT-USER2
Physical Config Desktop Programming Attributes
Command Prompt
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.11.4
Pinging 192.168.11.4 with 32 bytes of data:
Reply from 192.168.11.4: bytes=32 time=2ms TTL=125
Reply from 192.168.11.4: bytes=32 time=4ms TTL=125
Reply from 192.168.11.4: bytes=32 time=4ms TTL=125
Reply from 192.168.11.4: bytes=32 time=10ms TTL=125
Ping statistics for 192.168.11.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 10ms, Average = 5ms
C:\>

```



5.0 Ethical Consideration

To ensure a better security policy in the network, there should be a set of factors that should be put into consideration from the processes of designing, implementing and maintaining the DFS network. These aspects are very important in every business since they ensure better security that will protect you and your properties (Alharbi & Aspinall, 2018). The following are some of the major aspects that should be considered when designing, implementing and maintaining the DFS network.

a. Data protection laws

Every country has laws that regulate how data is handled from the processes of gathering, processing, and storage. Hence, the DFS network should be put into use in a manner that will adhere to all the British General Data Protection Regulation Act, and in any case, non-compliance attracts heavy penalties to the DFS's management. Therefore, the management should adhere to these legal issues in data sovereignty when designing, implementing and maintaining their network.

The proposed network has been designed and implemented in consideration to all the British General Data Protection Regulation Act.

b. Confidentiality Factors

This is a security principle that will ensure that the DFS network access and operability is free from any unauthorized persons. Getting unauthorized access to the network will be considered a violation of the confidentiality policy. Therefore, the network security should be able to provide adequate data protection mechanisms to prevent any third parties from

accessing the data communication. As suggested by SecureUD (2020), the manufacturers should implement data encryption to encrypt data and strong authentication mechanisms involving several steps, and only strong or recommended passwords are allowed. Cont et al. (2018) suggest that strong device authentication ensures that the connected devices function as expected.

In the proposed network, the confidentiality factor in the network has been implemented through the use of line console passwords that protects line console, enable password that protects privilege exec mode and IPsec VPN that enables secure communication between the HR department and the servers. Finally, the line console and enable passwords are encrypted to prevent attackers from seeing the plaintext passwords.

c. Integrity Factors

The privacy and data protection policy should be the key highlight of designing, implementing and maintaining the DFS network in order to ensure that the data communication is not modified by unauthorized persons. To prevent integrity violation, the devices should implement the use of public-key encryption for certificate-based digital signatures (Cont et al. 2018). The site-to-site IPsec VPN plays an important role in the network to ensure that all the data are encrypted and no unauthorized personnel can see or modify the data.

d. Availability Factors

While designing, implementing and maintaining the DFS network., the availability principle should be considered to ensure that the network provide uninterrupted services with no downtime in order to maintain the network system continuity and data (John, 2018). The implemented network security should be able to protect against threats such as Denial of service attacks. Also, the processes of data encryption and authentication should guarantee the availability of data.

In the proposed DFS network, we have ensured that availability factor is implemented through having redundancy in the network which is very important network design principle that offers to provide a backup path as an alternative to the primary path so that if the primary path fails, traffic will automatically run across the backup path hence network availability.

e. Data Storage and Cloud Computing Services

Data should be kept secured from the physical destruction or any other human-inflicted interference such as virus or hackers who may compromise the data (Ullah et al., 2019).

Among the security solutions to be implemented should include data centers and adopting cloud computing. The strategy will ensure that the services are move into the cloud, and the underlying infrastructure supports the operation of the DFS network.

In the proposed network, there is servers that are responsible with the storage of these users' information. The email server in the network will act to facilitating sending, receiving, processing and storage of mails in the network. For the security of data, site-to-site IPsec VPN is implemented to ensure that all the traffic between the HR department and the servers are encrypted and secured.

f. Data encryption

Missing data encryption compromises the information security of the CIA. For example, a code that is used to write users' logins to cookies so that they cannot log in again, store the username and passwords as plaintext hence the attacker who has access to the victim's machine can exploit these vulnerabilities and lead to more malicious activities.

To prevent these cases, users should employ cryptographic mechanisms to encrypt all the data they send and receive. A proper encryption mechanism will guarantee a near-bulletproof barrier against attackers and ensures that only authorized users are allowed to access resources.

In the proposed network, there is a secure tunnel that has been created between the HR department and the servers and hence all the data communication has been encrypted by the site-to-site IPsec VPN thus preventing any threats from attackers.

g. Authentication

Authentication methods such as dual-factor authentication should be enacted in the network as it is a security control in which two different authentication factors of users are required in verifying their identities for accessing a computer system or other resources: these different authentication factors are knowledge, possession, and inherence; hence this process will utilize the two of these factors.

In the proposed network, this authentication method will enhance security since it will add a significant layer of security in the process of authenticating users through making the process very complicated for the hackers or attackers to have access to user's accounts or devices because knowing the users' passwords by attackers alone will not be an efficient way in passing the authentication check in this security control. The attacker may have the user's password but not the user's second factor i.e., user's fingerprint

6.0 Conclusions

In conclusion, the proposed DFS network will ensure better performance, redundancy, scalability and security implementation principles to the overall company network and security systems when all the required expectations are put in place as explained in the above discussion. The proposed network will ensure that the company achieve a greater milestone in better market leads in terms of connecting the company to a more robust, reliable, high-performing and secured network. Therefore, based on our findings, we can recommend that the company should indeed implement the proposed network.